ANSWERS

1.

Question	Answer	Marks
5(a)(i)	3.2×10^{-19} / (–)1.6×10^{-19} **or** 2 more protons than electrons shown by charges	**B1**
	(2 + 2 =) 4	**B1**
5(a)(ii)	$^{9}_{4}Be$	**B1**
5(b)	a form of an element	**B1**
	with a particular number of neutrons / nucleons	**B1**
5(c)	electrons and nucleus opposite charged **or** electrons negative and nucleus positive	**B1**
	(electrostatic / electric) force of attraction **or** unlike charges attract	**B1**

2.

(a) left column both 1 B1
right column 0 **and** 1 B1

(b) (at least one of the atoms) contain same number of electrons and protons B1
or have 1 electron and 1 proton
charge on electron and proton opposite B1
or electron negative **and** proton positive
or charge on electron neutralises/cancels/balances proton charge
neutrons have no charge B1

3.

(a) $^{15}_{8}O$ /oxygen-15/oxygen (nucleus) B1

(b) (i) $^{12}_{6}C$ **and** $^{14}_{6}C$ /carbon-12 **and** carbon-14/the two carbon nuclei B1

(ii) $^{14}_{6}C$ **and** $^{14}_{7}N$ /carbon-14 **and** nitrogen-14 B1

(iii) $^{14}_{7}N$ **and** $^{15}_{8}O$ /nitrogen-14 **and** oxygen-14/the nitrogen and oxygen nuclei B1

4.

(a) (i) 3 **cao** B1

(ii) 208 **cao** B1

(iii) 11 **cao** B1

(b) (i) 17 **cao** B1

(ii) 20 **cao** B1